ModelIt!

Student Activity Pack

Building a Star on Earth: The Fusion Energy Challenge

Modeling the Engineering Extremes of Controlled Nuclear Fusion

Grade 10L3 | NGSS HS-ESS1-1, HS-PS3-5 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is nuclear fusion and how is it different from nuclear fission used in today's power plants?

2 Why do you think scientists have been trying to build a fusion reactor for over 70 years without commercial success?

3 Draw what you think happens inside a fusion reactor — where does the energy come from?

4 If fusion is what powers the Sun, why is it so hard to recreate on Earth?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Plasma Temperature | Magnetic Containment | Fuel Injection Rate | Energy Output | Neutron Damage | Cooling System | Power Conversion | Materials Degradation | Net Energy Gain

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: ITER Baseline

150 million degrees, 5.3 Tesla containment, moderate fuel injection

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Ignition Attempt

Maximum temperature and fuel injection

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Containment Instability

Magnetic field reduced by 20%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tritium Breeding Ratio — The rate at which the reactor blanket produces new tritium fuel from lithium — the reactor must breed more tritium than it consumes because natural tritium is extremely rare

Plasma Disruption Frequency — The rate of sudden plasma instabilities that dump enormous energy onto reactor walls — each disruption can cause significant structural damage

Superconductor Temperature — The operating temperature of high-temperature superconducting magnets, which must be maintained near -250 degrees Celsius even meters from 150-million-degree plasma

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why have scientists spent 70 years and billions of dollars trying to recreate what the Sun does — and why haven't they succeeded yet? Use evidence from your simulation to support your answers.

1. **How does your model illustrate the fundamental tension between maximizing fusion output and minimizing reactor damage?**

2. **Why is the Lawson criterion (temperature x density x time) such a demanding requirement to meet simultaneously?**

3. **What role does Materials Degradation play as a limiting factor on reactor operational lifetime?**

4. **How would your model change if a material were discovered that was immune to neutron damage?**

5. **What does your model reveal about why fusion has been 30 years away for the last 60 years?**

8 STEM Challenge: Design a Next-Generation Fusion Reactor Concept

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YOUR ENGINEERING MISSION

Design a fusion reactor concept that addresses the three biggest engineering bottlenecks: plasma confinement stability, neutron damage to materials, and net energy gain optimization.

THE SCENARIO

The Department of Energy has issued a Grand Challenge for high school students to propose innovative solutions to the three biggest obstacles to commercial fusion power. Your team must identify the most critical bottleneck in current fusion designs and propose an engineering approach that could break through it, using evidence from your computational model.

Engineering Design Thinking

1. WHAT CONFINEMENT APPROACH DOES YOUR REACTOR USE AND HOW DOES IT ADDRESS PLASMA INSTABILITY?

2. WHAT MATERIALS WOULD YOU PROPOSE FOR THE REACTOR BLANKET TO WITHSTAND NEUTRON BOMBARDMENT?

3. HOW DOES YOUR DESIGN ACHIEVE Q GREATER THAN 10 — WHAT SPECIFIC PARAMETERS ARE OPTIMIZED?

4. WHAT IS THE ESTIMATED TIMELINE AND COST FOR YOUR CONCEPT TO REACH COMMERCIAL VIABILITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Plasma Physicists study the behavior of superheated ionized gases and design confinement systems for fusion reactors. They work at national laboratories (PPPL, LLNL, ORNL), universities, and private fusion companies like Commonwealth Fusion Systems, TAE Technologies, and Helion Energy, earning $90,000-$180,000/year. Fusion is one of the fastest-growing career fields in energy science.